Q1 **ECONOMICS**
How should scarce GPU-hours be allocated for value net of carbon when teams can misreport their private value?

Q2 **COMPUTATION**
Can an exactly solved multi-team VCG beat first come, first served and stay auditable?

Q3 **BEHAVIOR**
Do people report truthfully and understand the rule? (Hugging Face Space, Section 4)

1 Strategic Reports from Teams

Teams report private value; demand and emissions are observable

Team	Demand d_i (h)	Reported r_i	CO_2e e_i (kg)	Score $s_i = r_i - \lambda e_i$
Team A	25	**9**	6.0	6.0
Team B	20	**6**	4.8	3.6
Team C	15	**3**	3.6	1.2
Team D	25	**9**	6.0	6.0
Team E	20	**6**	4.8	3.6
Team F	15	**3**	3.6	1.2

Information structure

INHERITED · Harsanyi 1968

- Private: project value v_i (the report r_i may differ)
- Observable: demand d_i and estimated emissions e_i
- Carbon weight $\lambda = 0.5$; capacity $K = 100$ GPU-hours
- Total demand = 120 GPU-hours > K (contested)

2 Carbon-Aware VCG Allocation

Select the feasible set maximizing carbon-adjusted reported value

Optimization problem

NEW · carbon-adjusted score

$$\text{maximize} \quad \sum_{i \in S} (r_i - \lambda e_i)$$

$$\text{subject to} \quad \sum_{i \in S} d_i \le K$$

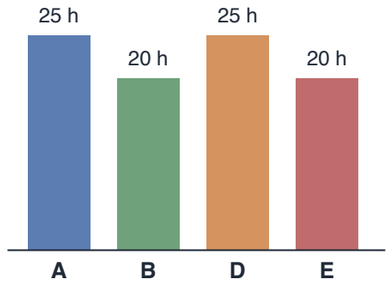
$K = 100$ GPU-hours, $\lambda = 0.5$

r_i : reported value
d_i : GPU-hours requested
e_i : estimated CO_2e (kg)
λ : carbon weight
K : capacity

Exhaustive search over feasible subsets

{A, B}
{A, C, D}
{B, C, E}
{A, B, D, E}
{A, D, F}
…

→ Winner set $S^* = \{A, B, D, E\}$



Total: 90 h ≤ K = 100 h

VCG payment for each winner i

INHERITED · Clarke–Groves

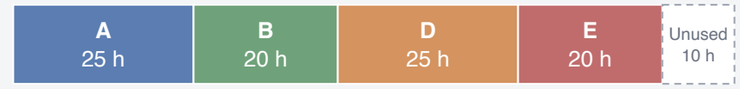
$$p_i = W^*_{-i} - \sum_{j \in S^* \setminus \{i\}} s_j$$

p_i : VCG payment of team i
W^*_{-i} : optimal value without i
s_j : carbon-adjusted score of j

3 Outcomes and Properties

Transparent allocation, payments, and carbon-aware efficiency

Resource allocation (K = 100 GPU-hours)



Payments (VCG)

24 priority credits each

Team A	24
Team B	24
Team D	24
Team E	24
Total	**96 credits**

Outcomes (selected set)

Total value Σv_i	**30**
Carbon-adjusted score	**19.2**
Total CO_2e (kg)	**21.6**
kg CO_2e per unit value	**0.72**

Comparison with first-come-first-served (FCFS)

FCFS score, illustrative single order	**15.6**
FCFS mean score over all 720 orders	**16.96**
Carbon-aware VCG score	**19.2**

✓ No FCFS order exceeds the VCG score

EVIDENCE · 10 checks

Key economic properties

Truthful reporting is dominant (DSIC)	Social welfare maximization	Carbon-aware and transparent
only under C1–C3 ↓		

Truthfulness holds only if

final credit design (Section 5)

C1 Own carbon charge
A winner pays λe in credits; otherwise the best report is v + λe

C2 Fixed credit budget
100 credits per team per round; no top-ups; spent credits retired

C3 Budget never binds
Largest spend 60 ≤ 100 credits over every λ and report (checked)

Intellectual merits

what we inherit, what is new, and what supports it

INHERITED
Vickrey 1961; Clarke 1971; Groves 1973: truthful prices
FAIR-Compute 2026: value-blind rules lose ~1/3 of value
Zhang et al. 2015: truthful cloud auction with costs
Radovanović et al. 2021: carbon-aware computing

NEW IN THIS WORK
Carbon priced into each team's score: s = r − λe
Credit design C1–C3 that makes the price costly
Exact comparison with all 720 FCFS arrival orders
Two GPU pools: same winners, 21.6 → 16.6 kg CO_2e

EVIDENCE
8 unit tests and 10 notebook checks pass (synthetic)
VCG 19.2 vs FCFS mean 16.96; no order beats VCG
0.72 vs 0.81–0.84 kg CO_2e per unit of value
n = 5 author self-plays: exploratory only